Noncanonical invariant hyperplanes below exponential order

Counterexamples for the family singled out in arXiv:2003.13573

Automatic Mathematics Run `fa_banach_001`, lane 11

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Status. Candidate counterexample, likely valid, pending human review. The proposed extension of the invariant-subspace classification fails for every subexponential power weight in both Hormander algebras and for the remaining infraexponential algebra at order one.

1 Source question

The source is J. Bonet and A. Galbis, *Invariant subspaces of the integration operators on Hörmander algebras and Korenblum type spaces*, arXiv:2003.13573 [1]. For a weight exponent p, the authors consider

$$A_p(\mathbb{C}) = \left\{ f \in H(\mathbb{C}) : \|f\|_{a,p} := \sup_{z\in\mathbb{C}} |f(z)|e^{-ap(z)} < \infty \text{ for some } a > 0 \right\}$$

with its locally convex inductive-limit topology, and

$$A_p^0(\mathbb{C}) = \left\{ f \in H(\mathbb{C}) : \|f\|_{a,p} < \infty \text{ for every } a > 0 \right\}$$

with its Frechet projective-limit topology. The integration operator is

$$\mathcal{J}f(z) := \int_0^z f(\zeta)\, d\zeta.$$

Theorem 5.1 classifies every proper closed $\mathcal{J}$-invariant subspace, under the hypothesis $p'(r) \to \infty$, as

$$A_K^E = \{f \in E : f^{(j)}(0) = 0,\ 0 \le j < K\}, \qquad K \in \mathbb{N}.$$

Remark 5.4, page 10, asks whether this conclusion remains true when the derivative hypothesis fails, particularly for $p(z) = |z|^s$, $0 < s \le 1$, except for the exponential-type algebra handled positively in Theorem 5.3.

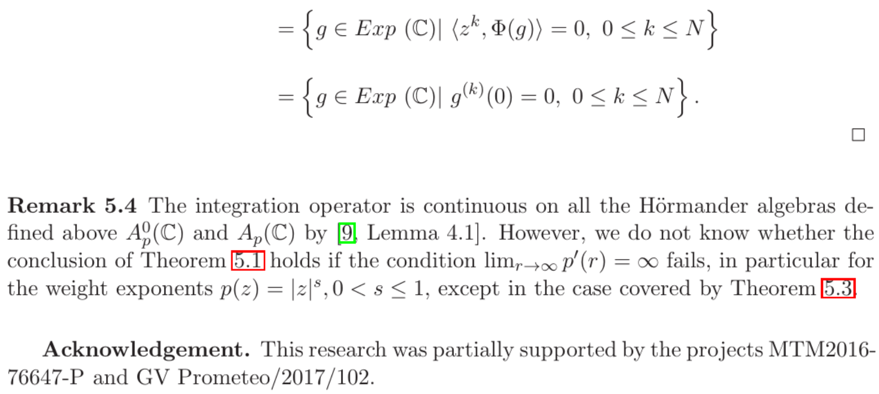
2 Proof intuition

Write an entire function in derivative coordinates $b_n = f^{(n)}(0)$. In these coordinates integration is the unilateral right shift:

$$(\mathcal{J}f)^{(0)}(0) = 0, \qquad (\mathcal{J}f)^{(n)}(0) = b_{n-1} \quad (n \ge 1).$$

Below exponential order, Cauchy's estimate forces (b_n) to decay faster than every geometric sequence. Consequently its generating function

$$\Lambda_\mu(f) = \sum_{n=0}^{\infty} \mu^n b_n$$



$$= \left\{ g \in Exp\ (\mathbb{C}) |\ \langle z^k, \Phi(g) \rangle = 0,\ 0 \le k \le N \right\}$$

$$= \left\{ g \in Exp\ (\mathbb{C}) |\ g^{(k)}(0) = 0,\ 0 \le k \le N \right\}.$$

□

Remark 5.4 The integration operator is continuous on all the Hörmander algebras defined above $A_p^0(\mathbb{C})$ and $A_p(\mathbb{C})$ by [9, Lemma 4.1]. However, we do not know whether the conclusion of Theorem 5.1 holds if the condition $\lim_{r\to\infty} p'(r) = \infty$ fails, in particular for the weight exponents $p(z) = |z|^s, 0 < s \le 1$, except in the case covered by Theorem 5.3.

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Figure 1: Remark 5.4 on page 10 of the source paper, together with the end of the positive exponential-type argument.

is a continuous linear functional for every $\mu \in \mathbb{C}$. It is an eigenvector of the transpose of $\mathcal{J}$, so its kernel is a closed invariant hyperplane. That hyperplane contains $z - \mu$, and therefore cannot be a vanishing-jet tail.

At order one the same construction works on the zero-type (infraexponential) Frechet algebra, because its topology supplies exponential seminorms of arbitrarily small type. It correctly stops working on the full exponential-type algebra: functions such as e^{az} make the derivative-generating series diverge when $|a\mu| \ge 1$. This is exactly the case already covered positively by the source paper.

3 Counterexample theorem

For $s > 0$, put $p_s(z) = |z|^s$, $A_s = A_{p_s}(\mathbb{C})$, and $A_s^0 = A_{p_s}^0(\mathbb{C})$.

Theorem 1. *Let $\mu \in \mathbb{C} \setminus \{0\}$.*

(i) *If $0 < s < 1$ and E is either A_s or A_s^0, then*

$$\Lambda_\mu(f) := \sum_{n=0}^{\infty} \mu^n f^{(n)}(0)$$

is a continuous nonzero linear functional on E.

(ii) *If $s = 1$ and $E = A_1^0$, the same conclusion holds.*

(iii) *In every case above,*

$$M_\mu := \ker \Lambda_\mu$$

is a proper closed codimension-one $\mathcal{J}$-invariant subspace of E which is not any canonical subspace A_K^E.

Moreover, the hyperplanes M_μ are pairwise distinct as μ ranges over $\mathbb{C} \setminus \{0\}$.

Thus the conclusion of Theorem 5.1 fails for every $0 < s < 1$ on both algebras named in Remark 5.4, and it fails for $s = 1$ on the remaining algebra A_1^0. The exception $A_1 = \mathrm{Exp}(\mathbb{C})$ is precisely the positive case of Theorem 5.3 in the source.

Proof. For $a > 0$, define

$$\|f\|_{a,s} := \sup_{z\in\mathbb{C}} |f(z)|e^{-a|z|^s}.$$

If this quantity is finite, Cauchy's estimate on the circle of radius $r > 0$ gives, for $n \geq 1$,

$$|f^{(n)}(0)| \leq n!\, r^{-n} \max_{|z|=r} |f(z)| \leq n!\, \|f\|_{a,s}\, e^{ar^s} r^{-n}.$$

The last factor is minimized when $asr^s = n$, hence

$$|f^{(n)}(0)| \leq n!\, \|f\|_{a,s} \left(\frac{eas}{n}\right)^{n/s} \leq \|f\|_{a,s} \left[(eas)^{1/s} n^{1-1/s}\right]^n, \tag{1}$$

where the second inequality uses $n! \leq n^n$. The $n = 0$ term is bounded by $\|f\|_{a,s}$.

Suppose first that $0 < s < 1$. Set $\delta = 1/s - 1 > 0$. From (1),

$$|\mu|^n |f^{(n)}(0)| \leq \|f\|_{a,s} \left[C_{a,s,\mu} n^{-\delta}\right]^n, \qquad C_{a,s,\mu} = |\mu|(eas)^{1/s}.$$

The scalar series $\sum_{n\geq 1} [C_{a,s,\mu} n^{-\delta}]^n$ converges. Therefore Λ_μ is absolutely defined and bounded with respect to $\|\cdot\|_{a,s}$ for every $a > 0$.

On A_s^0, any one of its defining seminorms now bounds Λ_μ, so the functional is continuous. The space A_s is the locally convex inductive limit of the normed growth steps used in the source paper. The restriction of Λ_μ to every step is continuous by the bound just proved; by the defining final topology of the inductive limit, Λ_μ is continuous on A_s as well.

Now let $s = 1$ and $E = A_1^0$. Given $\mu \neq 0$, choose $a > 0$ so small that $ea|\mu| < 1$. Formula (1) becomes

$$|\mu|^n |f^{(n)}(0)| \leq \|f\|_{a,1}\, (ea|\mu|)^n.$$

Since $\|\cdot\|_{a,1}$ is one of the defining seminorms of A_1^0, the series for Λ_μ converges absolutely and defines a continuous functional there.

In all cases $\Lambda_\mu(1) = 1$, so Λ_μ is nonzero. Direct differentiation of the defining integral gives

$$(\mathcal{J}f)^{(0)}(0) = 0, \qquad (\mathcal{J}f)^{(n)}(0) = f^{(n-1)}(0) \quad (n \geq 1).$$

Absolute convergence therefore permits the index shift

$$\Lambda_\mu(\mathcal{J}f) = \sum_{n=1}^{\infty} \mu^n f^{(n-1)}(0) = \mu\Lambda_\mu(f).$$

It follows that M_μ is a proper closed hyperplane and $\mathcal{J}(M_\mu) \subset M_\mu$.

Finally,

$$\Lambda_\mu(z - \mu) = \mu - \mu = 0,$$

while $(z - \mu)(0) = -\mu \neq 0$. Every proper canonical subspace A_K^E, $K \geq 1$, consists only of functions vanishing at zero. Thus M_μ is not of the form A_K^E. If $\nu \neq \mu$, then

$$\Lambda_\nu(z - \mu) = \nu - \mu \neq 0,$$

so $z - \mu \in M_\mu \setminus M_\nu$ and the hyperplanes are pairwise distinct. □

4 Verification and scope

The proof is analytic and uses no computational dependency. The companion verifier report redoes the optimization, the projective- and inductive-limit continuity arguments, the transpose eigenvalue identity, and the noncanonical-subspace test.

The theorem gives a full negative answer to the highlighted family in Remark 5.4 and disproves the universal extension suggested there. It does not classify all invariant subspaces for these algebras, nor does it decide which more general weight exponents with $p'(r) \not\to \infty$ admit similar eigenfunctionals. It also does not contradict the source's Theorem 5.3 for $\mathrm{Exp}(\mathbb{C})$, where Λ_μ is not defined on all functions.

5 Novelty check

A bounded web/arXiv search on 9 August 2026 used the source title and identifier, the combinations "integration operator invariant subspaces entire functions of order", "Hörmander algebras $p(z) = |z|^s$", "infraexponential invariant subspaces", and "Borel transform integration operator invariant subspace". The search returned the source paper and related Volterra/integration-operator papers, but no later arXiv paper explicitly resolving Remark 5.4 and no equivalent derivative-generating hyperplane construction. This supports, but cannot establish, novelty; confidence is moderate.

6 Human-review recommendation

The two points deserving focused review are: first, that continuity on each normed growth step gives continuity for the source's locally convex inductive-limit topology on A_s; second, that the source uses the unnormalized integral $\mathcal{J}f(z) = \int_0^z f$, which produces exactly the right shift in derivative coordinates. Subject to those checks, this should be accepted as a complete counterexample for every previously open power-weight case named in Remark 5.4.

References

[1] J. Bonet and A. Galbis, *Invariant subspaces of the integration operators on Hörmander algebras and Korenblum type spaces*, arXiv:2003.13573 (2020).